

2.4 Topic 4: Biodiversity and natural resources

Students will be assessed on their ability to:

- 1 Demonstrate knowledge and understanding of the practical and investigative skills identified in numbers 4 and 5 in the table of *How Science Works* on page 13 of this specification.
- 2 Compare the ultrastructure of plant cells (cell wall, chloroplasts, amyloplasts, vacuole, tonoplast, plasmodesmata, pits and middle lamella) with that of animal cells.
- 3 Compare the structure and function of the polysaccharides starch and cellulose including the role of hydrogen bonds between β -glucose molecules in the formation of cellulose microfibrils.
- 4 Explain how the arrangement of cellulose microfibrils in plant cell walls and secondary thickening contribute to the physical properties of plant fibres, which can be exploited by humans.
- 5 Compare the structures, position in the stem and function of sclerenchyma fibres (support) and xylem vessels (support and transport of water and mineral ions).
- 6 Describe how the uses of plant fibres and starch may contribute to sustainability, eg plant-based products to replace oil-based plastics.
- 7 Identify sclerenchyma fibres and xylem vessels as seen through a light microscope.
- 8 **Describe how to determine the tensile strength of plant fibres practically.**
- 9 Explain the importance of water and inorganic ions (nitrate, calcium ions and magnesium ions) to plants.
- 10 **Describe how to investigate plant mineral deficiencies practically.**
- 11 **Describe how to investigate the antimicrobial properties of plants.**

- 12 Compare historic drug testing with contemporary drug testing protocols, eg William Withering's digitalis soup; double blind trials; placebo; three-phased testing.
- 13 Explain the terms biodiversity and endemism and describe how biodiversity can be measured within a habitat using species richness and within a species using genetic diversity, eg variety of alleles in a gene pool.
- 14 Describe the concept of niche and discuss examples of adaptation of organisms to their environment (behavioural, physiological and anatomical).
- 15 Describe how natural selection can lead to adaptation and evolution.
- 16 Discuss the process and importance of critical evaluation of new data by the scientific community, which leads to new taxonomic groupings (ie three domains based on molecular phylogeny).
- 17 Discuss and evaluate the methods used by zoos and seedbanks in the conservation of endangered species and their genetic diversity (eg scientific research, captive breeding programmes, reintroduction programmes and education).